



Department of Computer Science and Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E Computer Science and Engineering

Course Code & Title: CB3491 Cryptography and Cyber Security

Name of the Faculty member (s): Mrs.K.Jeyakarthika, AP/CSE

Date of Implementation: 09.09.2023

Innovative Practice Description

- **Unit / Topic:** Unit III / Primality Testing
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO
- **Activity Chosen:** Flipped Classroom
- **Justification:**

Flipping the classroom is an inverting the classroom approach to teaching. In this approach, the traditional in-class teaching is “flipped” to better meet the needs of individual learners. Students gain control of the learning process through studying course material outside of class, using readings, pre-recorded video lectures. It helps the faculty/lecturer to redefine in-class activities and include homework problems and keep the students engaged in the content.

Time Allotted for the Activity: 20 minutes

Details of the Implementation

- **Plan:** Primality Testing - is given as self learning through self-exploration. Separate student's teams are formed and made them to explore as a group.
- **Identify and Share:** Related materials are identified and posted relevant references/materials/notes of primality testing to the students and given as self-learning. The Figure.1 shows the activities / discussion as a team on the self exploration topics.
- **Topic Allocation:** For the activity, teams were formed by their own interest. Each student in a group discusses among the peer members based on their self learning and write the points for Presentation within 10 minutes.
- **Topic Presentation:** Once they prepared the content for presentation with their peer members, each group present the given topic for maximum of 5 to 10 minute. The implementation of flipped class room activity as shown in figure1 and figure2.

CO – PO / PSO mapping:

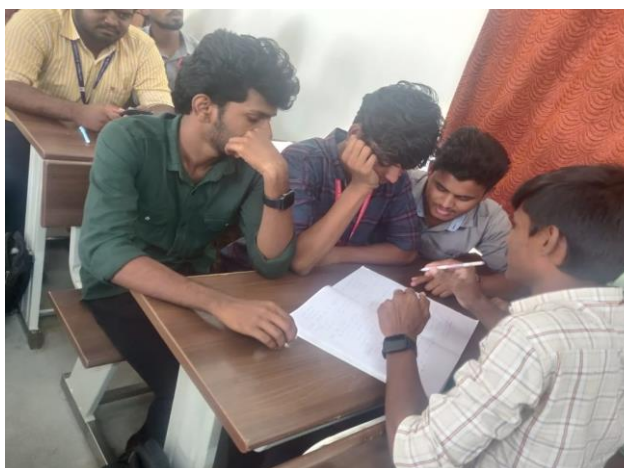
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO12
CO1	3	2	1	2	2	-	-	1	1	1
(1 – Low			2 – Moderate			3 – High)				



PO / PSO mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO12
	3	2	1	2	2	1	1	1
	Basic Computer Science Engineering and mathematics Knowledge for performing Asymmetric encryption is highly correlated	Analyzing various Asymmetric encryption algorithms is highly correlated	Developing Asymmetric encryption systems with basic encryption mechanisms are highly correlated	Investigate various aspects of cryptanalysis is highly correlated	Use of virtual environment and crypt tool for performing Asymmetric encryption is highly correlated.	The students will adhere to the ethics.	The students will work effectively as an individual and team with how to encrypt and decrypt a message using Asymmetric Encryption Algorithms.	Updating the Asymmetric encryption algorithms to the current trends are moderately correlated

• Images / Screenshot of the practice:





- **Reflective Critique:**

- ❖ Feedback of practice from students and other stakeholders:

- Student felt good, since they can study at their own pace/time.
- They felt that through such learning, they can explore more.

- ❖ Benefit of the practice:

- More one-to-one time between teacher and student.
- More collaboration time for students.
- Students learn at their own pace.
- Practical things – like missing class due to illness – become less problematic.
- It encourages students to come to class prepared.

- ❖ Challenges faced in implementation:

- Relies on student preparation – few students did not come prepared.
- The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a normal class.

References:

- ❖ <https://www.nyu.edu/faculty/teaching-and-learning-resources/strategies-for-teaching-with-tech/flipped-classes/implementing-a-flipped-class.html>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/2022-2023/CS8792-RV-Flipped-Unit-III.pdf>

Signature of Faculty Member

HOD-CSE